ReLaX: Reasoning with Latent Exploration for Large Reasoning Models

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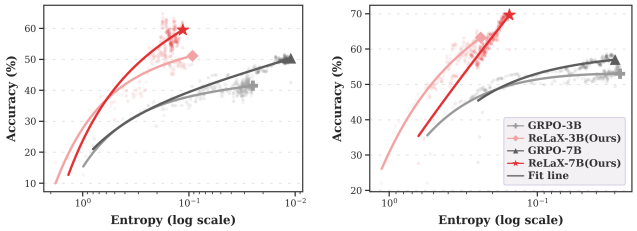
Abstract

Reinforcement Learning with Verifiable Rewards (RLVR) has recently demonstrated remarkable potential in enhancing the reasoning capability of Large Reasoning Models (LRMs). However, RLVR often leads to entropy collapse, resulting in premature policy convergence and performance saturation. While manipulating token-level entropy has proven effective for promoting policy exploration, we argue that the latent dynamics underlying token generation encode a far richer computational structure for steering policy optimization toward a more effective exploration–exploitation tradeoff. To enable tractable analysis and intervention of the latent dynamics of LRMs, we leverage Koopman operator theory to obtain a linearized representation of their hidden-state dynamics. This enables us to introduce ***D****ynamic* ***S****pectral* ***D****ispersion (****DSD****), a new metric to quantify the heterogeneity of the model's latent dynamics, serving as a direct indicator of policy exploration. Building upon these foundations, we propose* ***Re****asoning with* ***La****tent e****X****ploration (****ReLaX****), a paradigm that explicitly incorporates latent dynamics to regulate exploration and exploitation during policy optimization. Comprehensive experiments across a wide range of multimodal and text-only reasoning benchmarks show that ReLaX significantly mitigates premature convergence and consistently achieves state-of-the-art performance.*

1. Introduction

Scalable and verifiable reasoning stands as the foundational capability separating current foundation models from artificial general intelligence. Recent progress in RLVR [9, 13, 18, 30] has emerged as an effective paradigm for enhancing the capability of Large Language Models (LLMs) and Multimodal LLMs (MLLMs) on complex reasoning tasks. Whereas, without explicit intervention, reinforcement learning (RL) naturally drives a progressive reduction in policy entropy, which confines the policy gradient within a narrow subspace [1, 33]. The sparse reward in RLVR further exacerbates this *entropy collapse* problem, causing the policy to over-exploit prematurely, thereby inhibiting adequate exploration and ultimately leading to suboptimal performance. This bottleneck is supported by empirical evidence from Group Relative Policy Optimization (GRPO) [32, 51], which reveals an exponential relationship between policy entropy H and reward $\mathcal{R}$: $\mathcal{R} = -a \cdot \exp(H) + b$ [10] (see grey lines in Figure 1). To date, the community has widely recognized the exploration–exploitation tradeoff [35] as the fundamental challenge in scaling RL to achieve improved reasoning performance [11, 16, 22, 39, 44, 50].



(a) LLMs (Text-only) (b) MLLMs (Vision-language)

Figure 1. Empirical relationship between policy performance $\mathcal{R}$ and token-level entropy H during RLVR training with (a) text-only LLMs and (b) vision-language models (VLMs). Each scatter denotes a single training step, the solid curve fitted by $\mathcal{R} = -a \cdot \exp(H) + b$ [10].

Existing endeavors (with a comprehensive review in the Supplementary Material) mainly focus on token-level entropy, including reshaping the reward [6, 7, 16], redesigning policy objectives with entropy-based regularization [20, 47], and heuristically anchoring salient tokens to locally elevate the stochasticity [10, 26, 38, 42, 46, 48, 55]. Despite recent progress, the objective of maintaining higher token-level entropy inherently conflicts with the tendency of RL, which naturally gravitates toward deterministic, low-entropy policies. Moreover, mainstream MLLMs exhibit a pronounced misalignment between cross-modal internal computations and unimodal, text-centric outputs [2, 41], making it difficult for token-level feedback to accurately re-

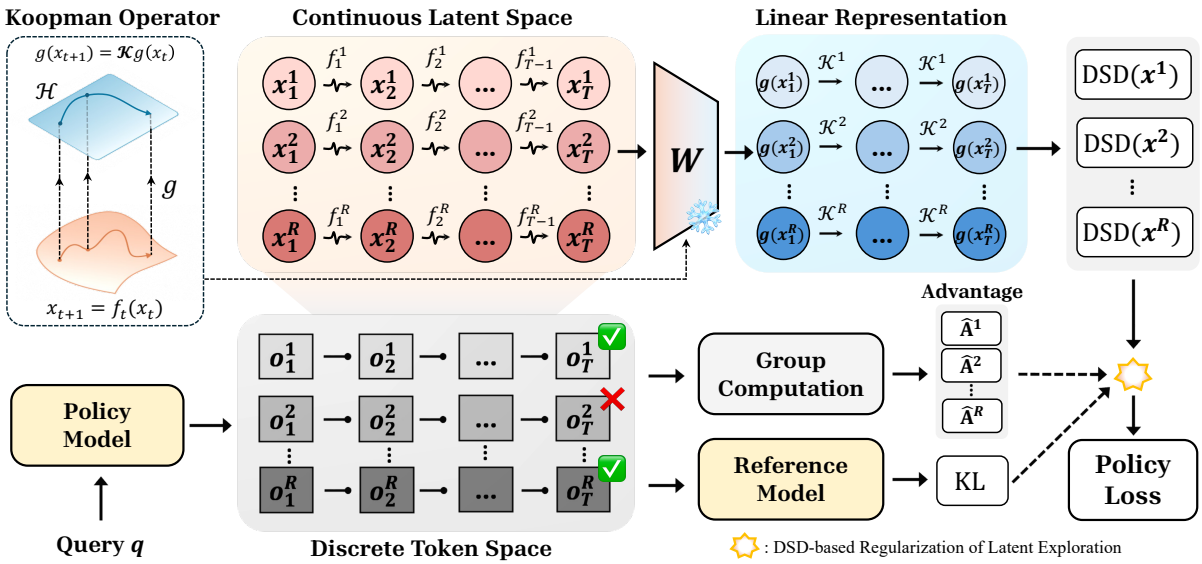


Figure 2. **Overview of ReLaX.** Grounded in Koopman operator theory (upper left), ReLaX employs a neural Koopman dictionary (frozen after one step of learning) during policy optimization to linearize the latent dynamics of last-layer hidden states. This transformation allows us to assess the flexibility of policy's internal computations through the proposed DSD. The DSD score for each trajectory is subsequently integrated into the GRPO objective, mitigating computational rigidity and enabling a more effective exploration–exploitation tradeoff.

flect the underlying multimodal processing. Taken together, these factors render existing methods inefficient and limit their generalizability when applied to MLLMs.

In this paper, we argue that entropy collapse is the superficial symptom of a deeper pathology: under RLVR, the internal computations that govern token generation gradually lose flexibility and converge into overly rigid patterns. These computations are instantiated through the latent dynamics—high-dimensional trajectories of hidden states that embed far richer and more stable inductive biases than what is observable in the discrete and sensitive token space. The key to harnessing the continuous latent dynamic of LRMs lies in finding an appropriate representation to make it analytically tractable. Modern Koopman operator theory [3] provides a powerful framework to represent the nonlinear dynamics of a model as linear evolution in an infinite-dimensional space of observables, with the Koopman dictionary governing the functional coordinates of this representation. Building upon this foundation, we introduce a novel metric, **Dynamic Spectral Dispersion (DSD)**, to characterize the flexibility of internal computations in LRMs by quantifying the degree of heterogeneity exhibited in their underlying latent dynamics. We then propose a latent dynamics aware paradigm for policy optimization, **ReLaX** (**Re**asoning with **La**tent e**X**ploration), that employs DSD-based regularization to counteract the rigid internal computation and facilitate effective exploration-exploitation tradeoff. Our main contributions are summarized as follows:

- We propose DSD, a metric that captures the heterogeneity of latent dynamics in LRMs, providing a more fundamental characterization of policy exploration by probing the model's internal computational processes rather than its surface-level token statistics.
- Incorporating DSD into policy optimization, we introduce a novel training paradigm, ReLaX, to effectively facilitate the exploration–exploitation tradeoff and mitigate the performance saturation of RLVR (see Figure 1).
- Extensive experimental results demonstrate that ReLaX substantially improves the capabilities of LRMs, setting new state-of-the-art results across 7 multimodal reasoning benchmarks and 6 text-only mathematical reasoning benchmarks.
- Comparative analyses with existing entropy-based methods showcase that ReLaX enables more robust and structured reasoning behaviors, while also generalizing more effectively to MLLMs. These findings indicate that ReLaX opens a promising direction for advancing reasoning capabilities by transcending token-space interventions and guiding exploration within the more expressive and computationally meaningful latent space.

2. Preliminary

2.1. Problem Formulation

The fine-tuning of foundation models for reasoning tasks can be formulated as a reinforcement learning problem driven by *verifiable rewards*. Let $\pi_\theta(o|q)$ denote the pol-

icy model parameterized by θ, which generates a reasoning trajectory $o = (o_0, o_1, \ldots, o_T)$ conditioned on a prompt $q \sim \mathcal{D}$, where $\mathcal{D}$ denotes the distribution of input prompts. The trajectory length is constrained by a maximum context length limit $T \leq L_{\max}$.

Each generated trajectory o is evaluated by a *verifier* that produces a scalar-valued $\mathrm{reward}(q, o)$. The verifier provides an objective and automatically checkable signal (e.g., correctness of a mathematical derivation or successful code execution), thus avoiding subjective biases and reward hacking issues in preference-based RL. The training objective of RLVR is to maximize the expected reward under the policy distribution:

$$\max_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\, o\sim\pi_\theta(\cdot|q)}\big[\mathrm{reward}(q, o)\big], \tag{1}$$

subject to $|o| \leq L_{\max}$.

2.2. Group Relative Policy Optimization

To efficiently optimize the RLVR objective, GRPO serves as a group-based variant of PPO, stabilizing policy improvement by normalizing rewards within each prompt group. For each prompt q, GRPO samples a group of R responses $\{o^i\}_{i=1}^r$ and estimates a group-relative advantage for each trajectory:

$$\hat{A}^i = \frac{\mathrm{reward}(q, o^i) - \mathrm{mean}\big[\mathrm{reward}(q, \{o^i\}_{i=1}^R)\big]}{\mathrm{std}\big[\mathrm{reward}(q, \{o^i\}_{i=1}^R)\big]}. \tag{2}$$

To handle off-policy data and constrain the optimization step, GRPO adopts the PPO-style clipped surrogate objective:

$$\begin{aligned}\mathcal{J}_{\mathrm{GRPO}}(\theta) \quad &= \mathbb{E}_{q\sim\mathcal{D},\{o^i\}_{i=1}^R\sim\pi_{\theta_{\mathrm{old}}}(\cdot|q)} \\ &\Bigg[\frac{1}{R}\sum_{i=1}^{R}\min\Bigg(\frac{\pi_\theta(o^i \mid q)}{\pi_{\theta_{\mathrm{old}}}(o^i \mid q)}A^i, \\ &\mathrm{clip}\Bigg(\frac{\pi_\theta(o^i \mid q)}{\pi_{\theta_{\mathrm{old}}}(o^i \mid q)}, 1-\epsilon, 1+\epsilon\Bigg)A^i\Bigg)\Bigg].\end{aligned} \tag{3}$$

To constrain policy updates and ensure stability, a KL-divergence regularization $\mathcal{L}_{\mathrm{KL}} = D_{\mathrm{KL}}(\pi_\theta \| \pi_{\mathrm{ref}})$ scaled by β is optionally applied to penalize large deviations from the initial policy. By leveraging group-relative normalization, GRPO provides a stable and scale-invariant learning signal, effectively reducing variance and improving sample efficiency in reasoning-oriented policy updates.

2.3. Koopman Operator Theory and Dynamic Mode Decomposition

The Koopman operator provides a rigorous framework for representing a nonlinear dynamical system within an infinite-dimensional Hilbert space $\mathcal{H}$. For a discrete-time system governed by $x_{t+1} = f_t(x_t)$, the Koopman operator $\mathcal{K}$ acts on observables $g \in \mathcal{H}$ (Koopman dictionary) as:

$$[\mathcal{K}g](x_t) := g(f_t(x_t)) = g(x_{t+1}). \tag{4}$$

Building upon Koopman operator theory, the Dynamic Mode Decomposition (DMD) [31] serves as a data-driven finite-dimensional approximation of $\mathcal{K}$. Consider a trajectory with consecutive states $\mathcal{V} = \{g(x_0), g(x_1) \ldots, g(x_{t-1})\}$ and their successors $\mathcal{V}^+ = \{g(x_1), g(x_2) \ldots, g(x_t)\}$, $\mathcal{K}$ can be estimated by solving a least-squares problem:

$$\mathcal{K} = \arg\min_{\mathcal{K}} \left\|\mathcal{V}^+ - \mathcal{K}\mathcal{V}\right\|_F^2 = \mathcal{V}^+\mathcal{V}^\dagger, \tag{5}$$

where $\mathcal{V}^\dagger$ denotes the pseudoinverse of $\mathcal{V}$.

Spectral analysis on $\mathcal{K}$ through its eigenfunctions and eigenvalues enables the characterization of underlying nonlinear dynamics. However, the discretization in DMD often introduces spurious eigenvalues when applied to complex systems with rich continuous spectra, leading to the loss of critical dynamical modes [25, 43]. Recent theoretical advances, particularly ResDMD [8], address this challenge by filtering out poorly convergent spectral components, thereby enabling a more accurate and reliable characterization of complex dynamics. This approach has also shown promise for analyzing the latent dynamics of LLMs [54].

3. Proposed Methodology

This section gives a comprehensive presentation of the proposed ReLaX. First, building on Koopman operator theory, Sec. 3.1 introduces DSD as a principled metric for capturing the heterogeneity of latent dynamics (Fig. 2, upper). Second, Sec. 3.2 describes how we we construct a reliable and accurate linear representation to support calculating DSD for LRMs. Finally, Sec. 3.3 details how DSD is incorporated into GRPO to achieve an effective exploration–exploitation tradeoff (Fig. 2, lower).

3.1. Dynamic Spectral Dispersion (DSD)

The explicit Chain-of-Thoughts (CoTs) in LRMs refer to probabilistic outcomes derived from continuous hidden representations that encapsulate fine-grained dynamical processes. However, the nonlinear and high-dimensional nature of latent dynamics presents substantial challenges for effectively and accurately capturing their underlying dynamics.

The Koopman operator provides a powerful framework for analyzing such nonlinear dynamics by transforming the original latent dynamics into a globally linear representation in a theoretical infinite-dimensional space. For each hidden state x, the corresponding Koopman operator $\mathcal{K}$

characterizes the linear temporal evolution of $g(x)$, whose spectral modes reveal the latent dynamics of growth, decay, and oscillation, ultimately shaping output reasoning patterns. Therefore, the distribution of these spectral modes can well reflect the flexibility of underlying computations during reasoning: *a concentrated spectrum denotes repetitive dynamics, while a dispersed spectrum marks richer dynamical regimes that yield more diverse outputs.*

Leveraging this insight, we put forward DSD as an operational proxy for the policy model's computational flexibility. Formally, given a sequence of hidden states $x \in \mathbb{R}^{T\times d}$ and its approximated Koopman operator $\mathcal{K}$, the DSD is defined as the variance of the Koopman eigenvalue magnitudes:

$$\mathrm{DSD}(x) = \mathrm{Var}(|\Lambda|), \text{ where } \mathcal{K}\Phi = \Phi\Lambda. \tag{6}$$

Intuitively, a higher DSD score signals a more expressive intrinsic dynamical spectrum, revealing a model whose internal computations remain flexible rather than collapsing into rigid patterns. Such flexibility endows the policy with a greater ability to explore novel trajectories throughout optimization.

3.2. Koopman Dictionary Learning

A core challenge in applying the Koopman operator is selecting an appropriate function space in which the operator $\mathcal{K}$ can faithfully linearize the policy model's latent dynamics—an especially difficult problem given the high dimensionality and strong nonlinearity of LRMs. To address this, we adopt ResKoopNet [45], an extension of ResDMD that learns a neural Koopman dictionary. This enables a more accurate and numerically stable approximation of the Koopman operator and its spectrum, which is essential for reliable DSD computation.

Specifically, the Koopman observables g are parameterized by a single linear layer W followed by a sigmoid activation $\sigma(\cdot)$:

$$g(x) = \sigma(Wx), \quad W \in \mathbb{R}^{d\times m}, \tag{7}$$

where m represents the dimensionality of the approximated Koopman operator. The dictionary W is optimized using hidden state trajectories $\{x_i\}_{i=1}^{B\times R}$ collected from the initial policy, where B denotes the policy training batch size. The optimization objective for W is to minimize the spectral residual of the Koopman operator:

$$W = \arg\min \frac{1}{BR} \left\| (\mathcal{V}^{+} - \mathcal{K}\mathcal{V})\Phi \right\|_F^2, \tag{8}$$

where Φ denotes the eigenvectors of $\mathcal{K}$. Once optimized, W is frozen during subsequent policy training to ensure a consistent function space for portraying policy latent dynamics throughout optimization. A more detailed presentation related to Koopman dictionary learning is provided in the Supplementary Material.

3.3. ReLaX

DSD characterizes the policy exploration from a computational perspective and supports gradient propagation, providing a principled basis for steering policy optimization toward an effective exploration–exploitation tradeoff. Formally, for hidden states $\{x^i\}_{i=1}^{R}$ correspond to group responses $\{o^i\}_{i=1}^{R}$, we define a sequence-level regularization term $\mathcal{L}_{\mathrm{xp}}$ associated with the corresponding DSD scores as:

$$\mathcal{L}_{\mathrm{xp}} = \log\left(\frac{1}{R}\sum_{i=1}^{R}\exp\big(-\mathrm{DSD}(x^i)\big)\right), \tag{9}$$

where the log-mean-exp computation smooths DSD numerically, enhancing gradient stability during optimization.

While this regularization alleviates computational rigidity by discouraging the latent dynamics from converging to a homogeneous mode, excessive exploration can undermine the necessary exploitation. Accordingly, ReLaX integrates two control mechanisms to maintain latent exploration at an appropriate operating level. First, to ensure that exploration occurs within a meaningful subspace of the latent dynamics, we weight the DSD scores by the advantages truncated to their positive part. This design constrains the policy model to become more flexible only along trajectories that yield positive reward, preventing uninformative or detrimental exploration. The resulting advantage-shaped regularization is expressed as:

$$\tilde{\mathcal{L}}_{\mathrm{xp}} = \log\left(\frac{1}{R}\sum_{i=1}^{R}\exp\big(-\mathrm{clip}(\hat{A}^i, 0)\cdot\mathrm{DSD}(x^i)\big)\right). \tag{10}$$

Nevertheless, an over-dispersion of dynamic spectral modes within positive trajectories still induces instability. KL regularization acts as an elastic constraint to stabilize training, albeit at the cost of slower convergence [10, 46, 48, 55]. We compormisely apply an adaptive KL regularization to softly constrain policy updates for trajectories that exhibit excessive dynamic divergence, while allowing those with remaining exploration potential to proceed freely. The overall objective of ReLaX is thus formulated as:

$$\mathcal{J}(\theta) = \mathcal{J}_{\mathrm{GRPO}}(\theta) + \alpha\,\tilde{\mathcal{L}}_{\mathrm{xp}} + \beta\sum_{i}^{\mathcal{I}} D_{\mathrm{KL}}\big(\pi_\theta(o^i)\,\|\,\pi_{\mathrm{ref}}(o^i)\big), \tag{11}$$

where α controls the strength of regularization for latent exploration, and $\mathcal{I} = \{\, i \mid \mathrm{DSD}(x^i) > \xi \,\}$ denotes the subset of trajectories whose DSD scores exceed a threshold ξ, thereby requiring KL penalization. A complete pseudocode of ReLaX is included in the Supplementary Material.

4. Experiments

To establish the effectiveness of ReLaX, we perform extensive experiments on both VLMs and LLMs at the 3B and

Model	MathVista *testmini*	MathVerse *testmini*	MathVision *test*	DynaMath *overall*	MMMU *val*	MMStar *overall*	EMMA *overall*	Average
General Multimodal LLMs								
Qwen2-VL-7B [41]	58.2	19.7	16.3	42.1	54.1	60.7	20.2	38.8
Qwen2.5-VL-3B [2]	62.3	33.5†	21.2	40.0†	46.3	55.9	19.2†	39.8
Qwen2.5-VL-7B [2]	68.2	49.2	25.1	53.2	54.3	63.9	21.5	47.9
Qwen2.5-VL-72B [2]	74.8	57.6	38.1	-	70.2	70.8	-	-
Intern2-VL-8B [5]	58.3	22.8	17.4	39.7	51.2	61.5	19.8	38.7
Intern2.5-VL-8B [5]	64.4	39.5	19.7	-	56.0	-	20.6	-
Llava-OV-7B [23]	63.2	26.2	18.5	-	48.8	-	18.3	-
Kimi-VL-16B [36]	68.7	44.9	21.4	-	55.7	-	-	-
Reasoning Multimodal LLMs								
MM-Eureka-7B [28]	73.0	50.3	26.9	54.4†	55.2	64.3†	23.5	49.7
MM-Eureka-8B [28]	67.1	40.4	22.2	-	49.2	-	21.5	-
Vision-R1-7B [17]	73.5	52.4	27.2	52.0†	54.7	60.9†	22.4	49.0
R1-VL-7B [52]	63.5	40.0	24.7	45.2	44.5	60.0	8.3	40.9
OpenVLThinker-7B [12]	70.2	47.9	25.3	38.6†	52.5	56.3†	26.6	45.3
VL-Rethinker-7B [37]	74.9	54.2	32.3	55.2†	56.7	64.2†	29.7	52.5
SRPO-7B [19]	75.8	55.8	32.9	-	57.1	-	29.6	-
ReLaX-VL-3B (Ours)	70.7	46.2	27.6	52.2	52.2	60.7	26.9	**48.1**
ReLaX-VL-7B (Ours)	77.1	55.7	30.2	55.9	57.4	65.5	30.6	**53.2**

Table 1. Comparison of VLM performance (mean@1 accuracy) across multiple multimodal reasoning benchmarks. For 7B scale LRMs, the top-performing and runner-up results of VLMs within each column are marked in red and blue, respectively. † indicates our reproduced results using publicly available models and standard evaluation code. "–" denotes missing results due to unavailable models.

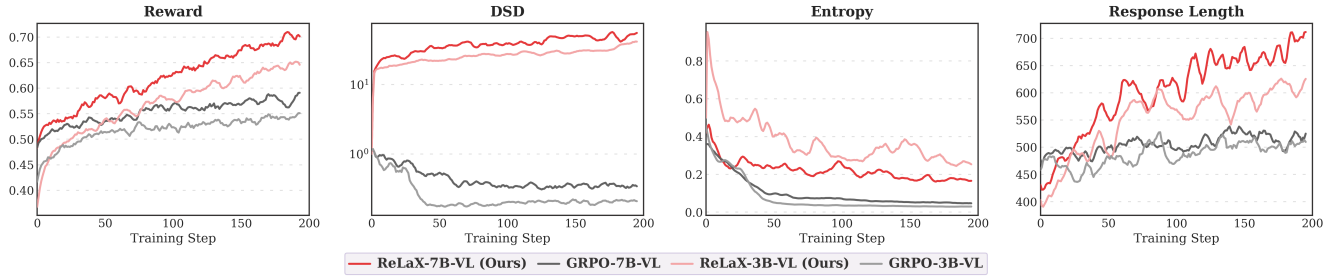


Figure 3. Comparison of training dynamics for *Reward*, *DSD*, *Entropy*, and *Response Length* under ReLaX (red) and vanilla GRPO (gray) on Qwen2.5-VL-Instruct at the 3B and 7B scales.

7B scales. We begin in Section 4.1 by outlining some important experimental setups. Section 4.2 then reports results on a broad suite of multimodal and text-only mathematical reasoning benchmarks, comparing ReLaX against state-of-the-art LRMs. Section 4.3 follows with an ablation study that illustrates how ReLaX attains a favorable exploration–exploitation tradeoff. Finally, beyond the benchmarking results, Section 4.4 offers a comprehensive comparison with prior token-level entropy–based methods.

4.1. Experimental Settings

We conduct experiments on both VLMs and LLMs within the GRPO framework, which is widely adopted in recent studies on the emergence of LRMs. Below, we outline the key experimental configurations used in our experiments:

4.1.1. Training Data and Benchmarks

VLMs are trained on the ViRL39K dataset from [37], which contains 38,870 curated and verifiable multimodal question–answer pairs. We adjust the query special tokens to align with our implementation framework. To comprehen-

Model	Size	MATH500	Minerva	AMC22	AMC23	AIME24	AIME25	Average
		Mean@1	*Mean@1*	*Mean@32*	*Mean@32*	*Mean@32*	*Mean@32*	
Qwen2.5-base	3B	18.4	2.6	12.5†	7.0	0.6	0.3	5.9
+Vanilla GRPO[32]	3B	69.2	23.5†	25.9†	51.6	7.5	4.5	26.0
+DCPO[46]	3B	71.2	-	-	55.8	7.5	4.7	-
+ReLaX (Ours)	3B	71.6	26.1	44.8	63.5	8.8	6.2	**31.6**
Qwen2.5-base	7B	64.6	5.7	21.4†	30.0	0.3	0.1	17.4
+SimpleRL[51]	7B	78.2	38.6	39.5†	62.5	15.6	8.9†	34.8
+DAPO[48]	7B	77.8	35.3	-	60.0	18.1	11.5	-
+KL-Cov[10]	7B	80.8	38.2	-	61.4	22.6	12.9	-
+HICRA[38]	7B	80.2	38.6	-	55.1	18.8	14.8	-
+FR3E[55]	7B	79.0	39.0	49.2†	67.5	25.2	14.8†	39.2
+ReLaX (Ours)	7B	82.4	39.1	65.4	84.1	19.7	13.8	**43.5**
Qwen2.5-Math-base	7B	52.4	10.7	25.4†	52.2	16.6	6.3	23.4
+SimpleRL[51]	7B	77.4	32.0	57.0†	60.8	26.7	9.3	37.6
+DAPO[48]	7B	81.6	38.2	-	62.5	31.6	14.9	-
+DCPO[46]	7B	82.5	-	-	79.8	38.8	17.2	-
+R1-zero-Div[47]	7B	77.2	31.6†	50.2†	58.0†	17.5†	10.0†	34.9
+FR3E[55]	7B	82.2	40.8	64.1†	67.5	26.7	18.1†	42.8
+ReLaX (Ours)	7B	85.6	43.4	71.9	88.9	36.9	17.3	**49.1**

Table 2. Comparison of LLM performance (mean@1 & mean@32) across multiple text-only mathematical reasoning benchmarks. For each base model, the top-performing and runner-up results of RLVR algorithms within each column are marked in red and blue, respectively. † indicates our reproduced results using publicly available models and standard evaluation code. "–" denotes missing results due to unavailable models.

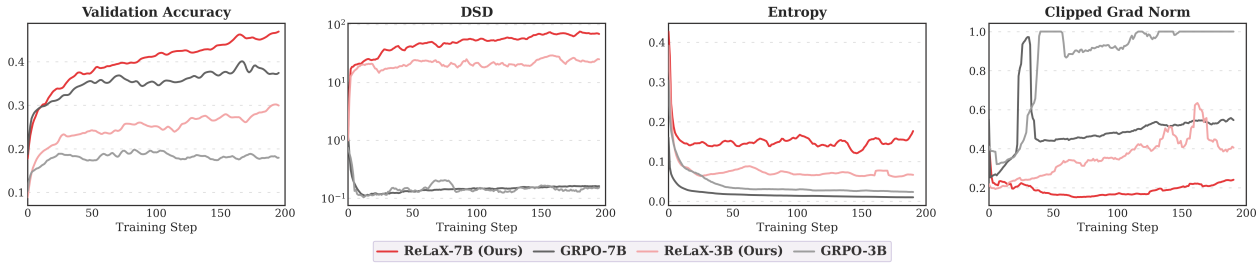


Figure 4. Comparison of training dynamics for *Validation Accuracy*, *DSD*, *Entropy*, and *Clipped Gradient Norm* under ReLaX (red) and vanilla GRPO (gray) on Qwen2.5-Base at the 3B and 7B scales.

sively evaluate multimodal reasoning performance, 7 challenging multimodal reasoning benchmarks are involved, including multidisciplinary datasets (MMMU [49], MM-Star [4], EMMA [14]) and mathematical datasets (MathVista [27], MathVerse [53], MathVision [40], and Dyna-Math [56]). We report the mean@1 accuracy using greedy decoding.

For LLMs training, we construct a merged dataset by combining the DAPO-Math-17K corpus [48] with the Level 3–5 subsets of the MATH training set [15], yielding approximately 22K mathematical queries with a broader range of difficulty. For evaluation, we use MATH500 (500 random samples from the MATH test set), Minerva [21], AMC 2022 & 2023 [29], and AIME 2024 & 2025 [24]. We report mean@1 on MATH500 and Minerva, and mean@32 on AMC and AIME for robust evaluation due to the limited size of these test sets.

4.1.2. Baselines and Implementations

The multimodal baselines include MM-Eureka [28], Vision-R1 [17], R1-VL [52], OpenVLThinker [12], VL-Rethinker [37], and SRPO [19]. For text-only mathematical reasoning, we benchmark ReLaX against a set of recent RLVR algorithms, including SimpleRL [51], DAPO [46],

KL-Cov [10], R1-zero-Div [47], and FR3E [55]. Our method is implemented on the VeRL codebase [34]. Additional details on baselines, training hyperparameters, evaluation settings, and our ReLaX implementation are provided in the Supplementary Material.

4.2. Main Results

4.2.1. Multimodal Reasoning

For multimodal experiments, we adopt Qwen2.5-VL-Instruct as the base model. As shown in Table 1, ReLaX-VL-3B and ReLaX-VL-7B yield absolute improvements of 8.3 and 5.3 in average mean@1 accuracy across 7 multimodal benchmarks, respectively, over their corresponding base models. Notably, ReLaX-VL-7B achieves an average score of 53.2, establishing a new state-of-the-art among existing 7B scale multimodal reasoning models and surpassing the previous best, VL-Rethinker-7B (52.5). At the 3B scale, ReLaX-VL-3B also shows strong competitiveness, outperforming several previous 7B-level models, including R1-VL (40.9) and OpenVLThinker (45.3).

The training dynamics shown in Fig. 3 further highlight how ReLaX enhances the capability of multimodal LRMs. Under vanilla GRPO (gray), both policy entropy and DSD exhibit a rapid decline within the first 50 steps. This indicates that the model quickly collapses into a rigid pattern—both in its internal latent dynamics and output tokens—ultimately leading to stagnant policy improvement due to reduced flexibility. In contrast, ReLaX maintains substantially more diverse latent dynamics and stabilizes the entropy at a higher yet well-regulated level. This balanced behavior enables the policy to continue improving and ultimately achieves a relative performance gain of 10% on both the 3B and 7B models. These results demonstrate that ReLaX unlocks a substantial amount of previously untapped potential by explicitly facilitating the exploration–exploitation tradeoff during RLVR—an aspect that prior MLLM work has largely overlooked.

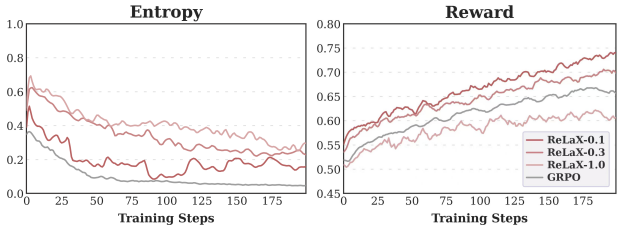
4.2.2. Text-only Mathematical Reasoning

For text-only reasoning, we conduct experiments on three base models: Qwen2.5-3B-Base, Qwen2.5-7B-Base, and Qwen2.5-7B-Math. We compare ReLaX against existing RLVR algorithms, including the GRPO baseline (referred to as SimpleRL [51] for 7B-scale models) and several GRPO variants designed to promote the policy exploration by manipulating token-level entropy. As shown in Table 2, ReLaX achieves substantial improvements over all three base models as well as their vanilla GRPO counterparts. Moreover, ReLaX consistently outperforms existing methods based on token-level entropy across 6 mathematical reasoning benchmarks. In particular, ReLaX surpasses the previous state-of-the-art publicly available baseline, FR3E [55], by 4.3 on Qwen2.5-7B-Base and 6.3 on Qwen2.5-7B-Math. To showcase that ReLaX generalizes across model families, we additionally evaluate it on Llama3.2-3B-Instruct and Qwen3-4B. The corresponding results are provided in the Supplementary Material.

We observe training dynamics in the text-only scenario that mirror the advantages seen previously in the multimodal experiments. Compared with vanilla GRPO, ReLaX maintains higher policy entropy, more diverse latent dynamics, and longer response lengths. Moreover, ReLaX exhibits strong training stability, as evidenced by its consistently low gradient clipping rate. Together, these properties enable ReLaX to achieve substantial improvements in both policy reward and validation accuracy.

4.3. Ablation Study

To understand the sources of ReLaX's superior performance, we conduct ablation studies on three key designs in DSD-based regularization that collectively enable an effective exploration–exploitation tradeoff.



(a) (b)

Figure 5. Training dynamics of policy entropy and reward on 3B scale Qwen2.5-VL-3B models by ReLaX with different DSD-based regularization coefficients (1.0, 0.3, 0.1, 0).

As shown in Fig. 5, we first evaluate the impact of the DSD-based regularization coefficient, which governs the strength of encouraging the policy to explore more diverse latent dynamics. Increasing this coefficient leads to a pronounced rise in policy entropy, indicating that nudging the model away from rigid internal computation regimes effectively mitigates entropy collapse. However, higher entropy does not necessarily translate into better policy convergence. We observe that ReLaX achieves the highest policy reward when the coefficient is set to 0.1. A coefficient of 0.3 yields only a marginal improvement over vanilla GRPO, while an overly strong coefficient (e.g., 1.0) harms convergence. Overall, these findings reveal that ReLaX benefits most from controlled rather than aggressive latent exploration, highlighting the importance of tuning α to maintain a productive learning regime.

Beyond adjusting the regularization strength via the coefficient, ReLaX further incorporates two mechanisms—adaptive KL regularization and advantage shaping—to stabilize the policy optimization. To assess their contributions, we remove these components and conduct

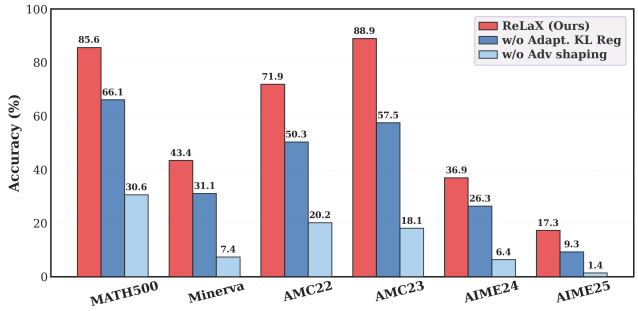


Figure 6. Evaluation results from the ablation study on Qwen2.5-7B-Math for text-only reasoning tasks. Results by full ReLaX is highlighted in red, while the dark-blue and light-blue bars respectively correspond to its ablations without adaptive KL regularization and advantage shaping.

ablation experiments on Qwen2.5-7B-Math and evaluate the performance across 6 mathematical reasoning benchmarks. As shown in Fig. 6, removing adaptive KL regularization causes the penalty to be applied uniformly across all sampled trajectories, regardless of their DSD scores. This indiscriminate constraint leads to consistently suboptimal performance across benchmarks, echoing observations from prior studies [10, 48] that KL regularization can inadvertently suppress useful policy updates. In contrast, ReLaX applies KL regularization adaptively by penalizing only trajectories exhibiting overly heterogeneous internal dynamics. Strikingly, removing advantage shaping severely degrades performance. Without this mechanism, exploration is promoted indiscriminately in latent space, including directions associated with negative or uninformative rewards. This leads to a collapse in policy quality, in some cases performing worse than the base model itself. The result underscores a key insight that effective exploration must be conditional—directed toward trajectories that are demonstrably beneficial, rather than uniformly expanding the search space.

4.4. Comparative Analysis

The core idea of ReLaX is to move beyond token-level entropy and instead leverage the richer structure encoded in a model's latent dynamics. Unlike prior approaches that regulate uncertainty solely in token space, ReLaX operates directly on the internal computational trajectories of the policy model. In this section, we present additional experiments that compare ReLaX with representative existing methods to highlight the benefits of this latent dynamics perspective.

4.4.1. Multimodal Generalization

As previously shown in Table. 2, ReLaX outperforms existing methods that mitigate entropy collapse of text-only LRMs by directly boosting their token-level entropy. On the other hand for MLLMs, mainstream architectures exhibit an apparant input–output modality misalignment: cross-modal computation is predominantly carried out in the latent space, not in the text-centric token space. To this end, entropy-based methods that operate solely on token distributions are intrinsically limited in their ability to shape multimodal processing, leaving their effectiveness in multimodal settings fundamentally constrained. To examine this, we evaluate entropy regularization [47] and KL-Cov [10] on Qwen2.5-VL-3B using the same training setting.

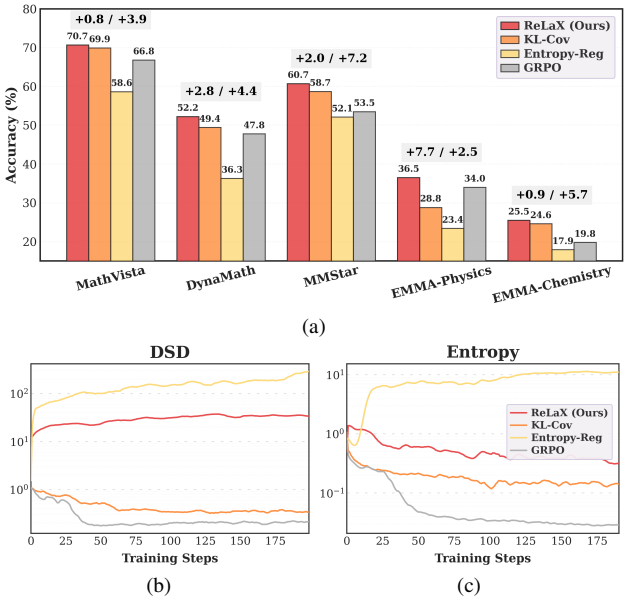


Figure 7. Comparison of RLVR training methods on Qwen2.5-3B-VL. (a) Evaluation results across 5 multimodal reasoning benchmarks. The values above each bar denote ReLaX's performance gains over KL-Cov (left) and vanilla GRPO (right). (b) and (c) Training dynamics of policy DSD and policy entropy, respectively.

As shown in Fig. 7, ReLaX consistently outperforms both entropy regularization and KL-Cov across multimodal benchmarks. Naively extending entropy regularization to multimodal training indeed increases both DSD and token-level entropy, this boost comes at the cost of highly chaotic latent computations, which in turn drift output semantic and significantly degrade performance. KL-Cov provides modest improvements; however, its training dynamics show only a mild increase in the heterogeneity of latent dynamics, indicating that token-level perturbations exert limited influence on the rigid cross-modal computational pathways of the policy model.

Interestingly, KL-Cov shows comparatively stronger gains on multimodal mathematics benchmarks such as MathVista, MathVerse, and MathVision (complete results in the Supplementary Material), where reliance on the visual modality is relatively limited. In contrast, on more vi-

sually grounded and discipline-rich tasks, ReLaX achieves substantially larger improvements—for instance, a 7.7 accuracy gain on EMMA-Physics. These results demonstrate that replacing token-level perturbations with mechanisms that explicitly promote diverse latent dynamics enables ReLaX to better generalize in multimodal reasoning.

4.4.2. Qualitative Results

To provide more intuitive evidence, we further present qualitative case studies that illustrate how ReLaX enhances the reasoning behaviors of LRMs beyond their quantitative benchmark gains, while also highlighting the disadvantages and unintended side effects introduced by simply increasing token-level entropy.

Firstly, we compare ReLaX with KL-Cov on the Qwen2.5-VL-3B model using a sample from the DynaMath dataset [56], which evaluates reasoning robustness by testing whether models remain consistent across variants of the same question (e.g., modified numerical values or visual details). As shown in Supplementary Material Tab. 7, ReLaX consistently interprets the visual context and applies the correct solution steps across all variants. In contrast, although KL-Cov answers the original question correctly, it fails on several variants—either by misinterpreting altered visual cues or by following incorrect solution procedures.

We also examine text-only reasoning using a query from AMC23 [29], comparing ReLaX-7B with the publicly released R1-zero-Div [47] model trained via entropy regularization. As shown in Tab. 9, both models obtain the correct final answer, yet their reasoning behaviors differ fundamentally. ReLaX-7B performs meaningful self-validation by invoking relevant mathematical principles. In contrast, R1-zero-Div attempts to "verify" its answer by generating Python code—an invalid and hallucinated strategy, as the model cannot execute code and thus cannot benefit from this verification attempt.

These results highlight an important insight: increasing token-level perturbations does not guarantee meaningful exploration and may even induce pathological or hallucinated behaviors. In contrast, by enhancing the flexibility of the model's internal computation, ReLaX fosters more robust and structured reasoning processes, achieving a more favorable balance between diversity and correctness. Additional case studies and detailed analysis can be found in the Supplementary Material.

5. Conclusion

In this work, we introduced ReLaX, a new RLVR training paradigm that explicitly promotes heterogeneity in latent dynamics during policy optimization. By enabling more flexible internal computations, ReLaX effectively mitigates the performance saturation induced by mode collapse—a fundamental bottleneck in RLVR for LRMs. Extensive evaluations across both multimodal and text-only reasoning benchmarks demonstrate that ReLaX substantially strengthens the reasoning capabilities of diverse model families. Moreover, comprehensive analyses show that ReLaX elicits more robust and structured reasoning behaviors, along with substantially improved generalization in multimodal settings, outperforming approaches that focus on increasing token-level entropy. These findings indicate that steering exploration directly within the latent space offers a more stable and principled mechanism for managing the exploration–exploitation tradeoff. Taken together, this study underscores the promise of harnessing the informative latent space as a powerful and scalable pathway for advancing the capabilities of LRMs.

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ReLaX: Reasoning with Latent Exploration for Large Reasoning Models

Supplementary Material

1. Overview

This supplementary material provides more details and results of the proposed ReLaX. In Supp. Sec. 2, we discuss the **related work** of recent GRPO variants that aim to address the RLVR's performance bottleneck through the lens of token-level entropy. In Supp. Sec. 3, we provide supplementary background on **Koopman dictionary learning** and the **pseudocode** of ReLaX. We further comprehensively describe our **experimental setup** in Supp. Sec. 4. Finally, **additional experimental results** are provided in Supp. Sec. 5 to complement the analyses in Sec. 4.

2. Related Work

Although RLVR has achieved impressive gains in mathematical, coding, and visual reasoning tasks, recent studies [44, 50] suggest that its improvements largely arise from enhanced sampling efficiency rather than true capability gains. What is worse, as the model becomes increasingly confident and generates more convergent outputs, policy learning suffers from reduced exploration, leading to saturation and a performance bottleneck—a central challenge in balancing exploration and exploitation [10, 33]. To mitigate this issue, recent research focuses on maintaining sufficient entropy in the model's action space to promote effective exploration.

Early methods [20, 47] seek to maximize policy entropy directly or incorporate it into the reward signal [7]. Such methods, however, require careful tuning of weighting coefficients: insufficient weighting fails to prevent collapse, while excessive weighting may induce semantic drift.

To regulate policy entropy in a more principled and controllable way, some research focuses on identifying structural factors that drive changes in entropy. DAPO [48] observes that the standard PPO/GRPO clipping range, though stabilizing the training, imposes an overly restrictive upper bound that suppresses exploration. Relaxing this bound via the clip-higher strategy increases credit for positively rewarded tokens and promotes broader exploration. Building on this insight, DCPO [46] introduces a dynamic, adaptive clipping mechanism that affords finer control over update magnitudes. Cui et al. [10] further reveal that the covariance between action probabilities and logit variations strongly predicts entropy reduction; accordingly, they propose KL-Cov and Clip-Cov to selectively penalize or clip high-covariance tokens most likely to induce entropy collapse.

Another line of approaches focuses on allocating policy-update credit selectively to high-entropy tokens. For instance, [42] updates only the top 20% high-entropy forking tokens and reports substantial performance gains. FR3E [55] extends this idea by identifying high-entropy tokens during sampling and expanding rollouts specifically along these positions; subsequent policy optimization is then performed on these augmented trajectories, effectively enhancing exploration. CURE [26] further generalizes FR3E by stochastically selecting high-entropy tokens to mitigate selection bias. To consolidate the improved exploration into stronger exploitation, CURE additionally applies DAPO [48] in a second training stage.

3. Details on Koopman Dictionary Learning

This section provides background and details the procedure for learning the Koopman dictionary, which maps the model's nonlinear, high-dimensional hidden states into a linear, tractable representation. The pseudocode for dictionary learning and the complete ReLaX training workflow is given in Algorithm 1.

3.1. Residual DMD (ResDMD)

Approximating the Koopman operator requires projecting its infinite-dimensional dynamics onto a finite set of observables [25]. This projection inevitably introduces spurious eigenvalues, a problem exacerbated for large reasoning models whose hidden states are extremely high-dimensional (e.g., 3584 for Qwen2.5-7B) and evolve over long contexts. In such settings, standard DMD approaches overfit transient fluctuations, suffer numerical instability, and produce spectral artifacts that obscure the true temporal modes.

ResDMD [8] improves spectral fidelity by diagnosing and filtering corrupted eigenvalues using a residual test. Given snapshot matrices $\mathcal{V}$ and $\mathcal{V}^+$ and an approximate Koopman operator $\mathcal{K}$, the squared residual of an eigenpair (λ, v) is estimated as:

$$\begin{aligned}\operatorname{res}(\lambda, v)^2 := \; & v^* \big[(\mathcal{V}^+)^* \mathcal{V}^+ - \lambda (\mathcal{V}^* \mathcal{V}^+)^* - \bar{\lambda}\, \mathcal{V}^* \mathcal{V}^+ \\ & + |\lambda|^2\, \mathcal{V}^* \mathcal{V} \big] v. \end{aligned} \tag{12}$$

This residual measures how well the eigenpair satisfies the least-squares formulation and, therefore, how faithfully it captures the underlying dynamics. Eigenpairs with large residuals are classified as spurious and removed. A limitation, however, is that this diagnostic only becomes available once the Koopman spectrum is computed with a fixed set of observables. Since choosing an effective dictionary is itself a central and challenging problem, a natural question

Algorithm 1: ReLaX

Input: Dataset $\mathcal{D}$, policy π_θ, reference policy π_{ref}, batch size B, group size R, learning rate η, total training steps $\mathcal{S}$
Output: Optimized policy π_θ

Set iteration counter $s \leftarrow 0$
while $s \leq \mathcal{S}$ **do**
  Sample a batch of queries $Q \sim \mathcal{D}$
  Initialize gradient accumulator $\nabla \mathcal{J} \leftarrow 0$
  `/* 1. Sampling */`
  foreach $q \in Q$ **do**
    Generate R rollouts $O = \{o^1, \ldots, o^R\}$ from $\pi_{\theta_{\text{old}}}(\cdot \mid q)$
    Collect hidden states $X = \{x^1, \ldots, x^R\}$ from the final hidden layer
    Compute $\text{reward}(Q, O)$ for rollouts
  end
  `/* 2. Koopman Dictionary Learning (First step only) */`
  if $s = 0$ **then**
    Form batch-level hidden-state set: $X_B = \bigcup_{q \in Q} X^q$
    $g \leftarrow$ `FitKoopmanDict`(X_B)
  end
  `/* 3. Policy Optimization */`
  foreach $x \in X$ **do**
    Compute $\text{DSD}(x)$ using Eq. 6
  end
  Compute ReLaX objective $\mathcal{J}(\theta)$ using Eq. 11
  Update policy: $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{J}(\theta)$
  $s \leftarrow s + 1$
end

Function `FitKoopmanDict` *(X)*:
  Input: Hidden states $X \in \mathbb{R}^{BR \times T \times d}$
  Select batch size B_g, total optimization steps $\mathcal{S}_g$, Koopman dimension m, and learning rate η_g
  Initialize dictionary network g with parameters $W \in \mathbb{R}^{d \times m}$ and sigmoid activation σ
  Set iteration counter $s \leftarrow 0$
  while $s \leq \mathcal{S}_g$ **do**
    Sample hidden state batch $X_B \sim X$
    Construct consecutive temporal snapshots $\mathcal{V}_B, \mathcal{V}_B^+ \in \mathbb{R}^{B_g \times (T-1) \times d}$
    Estimate Koopman operator $\mathcal{K}$ using Eq. 5
    Compute residual loss $\mathcal{J}_g(W)$ using Eq. 8
    Update dictionary parameters $W \leftarrow W - \eta_g \nabla_W \mathcal{J}_g(W)$
    $s \leftarrow s + 1$
  end
  return *Learned Koopman dictionary* g

arises: can we learn the dictionary jointly while minimizing spectral residuals?

3.2. ResKoopNet

ResKoopNet [45] addresses this by parameterizing the Koopman dictionary with an MLP and optimizing it directly through the residual objective in Eq. 12. The resulting residual loss is given in Eq. 8, with the full derivation provided in the original paper.

In our implementation, the dictionary is parametrized by a single feedforward layer with a Sigmoid activation. It is trained on the hidden states collected during the first step of sampling, yielding a general set of observables that defines a globally shared linear representation space for the subsequent policy optimization. The complete procedure is provided in Algorithm 1.

4. Detailed Experimental Settings

Our experiments were run on multiple GPU clusters, each equipped with 8× NVIDIA A100 (80GB). To improve training efficiency, we enabled actor–learner collocation via VeRL [34]. The primary hyperparameters on finetuning both LLMs and VLMs, including those for the actor, trainer, and ReLaX-specific settings, are summarized in Table 3. All experiments employ identical configurations for optimizing the Koopman dictionary, using the Adam optimizer with a learning rate of 10^{-4} and a batch size of 64.

Parameter	**Value**
Actor	
Maximum response length	3072
Temperature	1.0
top p	1.0
top k	-1
Number of rollouts per prompt	16
Trainer	
Batch size	512
Mini batch size	32
Generate size for sampling	2048
Optimizer	AdamW
Adam betas	(0.9, 0.95)
Gradient norm clipping	1.0
Learning rate scheduler	Constant
Learning rate	10^{-6}
ReLaX-Specific	
Exploration coefficient α	0.1
KL loss coefficient β	0.01
DSD threshold ξ	25
Koopman operator dimension m	50

Table 3. Hyperparameters for RLVR training used in our experiments. The same settings are applied to both VLM and LLM experiments.

The benchmarking results reported in Tables 1 and 2 are compiled from published papers and our evaluations on publicly available checkpoints using open-source toolkits. Specifically, VLMs are evaluated with VLMEvalKit [1], and LLMs are evaluated following Qwen2.5-Math [2]. The evaluation hyperparameters for both VLMs and LLMs are listed in Table 4.

Finally, the chat templates employed in VeRL for model training are presented below. During evaluation, we use the evaluation codebase's default system prompt to ensure fair and comparable results.

[1]https://github.com/open-compass/VLMEvalKit
[2]https://github.com/QwenLM/Qwen2.5-Math

Parameter	**Value**
VLM evaluation	
top p	0.1
Maximum response length	10240
Temperature	0
LLM-as-Judge	GPT-4o-mini
LLM evaluation	
top p	0.1
Maximum response length	10240
Temperature (mean@1)	0
Temperature (mean@32)	1

Table 4. Hyperparameters used for Evaluations.

Chat template for Visual-Language training.

System: Given the images and the question, please reason step by step, and put your final answer within \boxed{}.
User: ⟨Question⟩
Assistant:

Chat template for text-only training.

System: Please reason step by step, and put your final answer within \boxed{}.
User: ⟨Question⟩
Assistant:

5. Additional Results

In this section, we present the additional results and details to clarify some of the analytical experiments and substantiate our claims.

5.1. Supplemented Experiments on Other Models

Although models from the Qwen2.5 family are commonly used as base models for RLVR training in recent work, there has been an ongoing community discussion regarding potential evaluation set leaks. To this end, we further include experiments on Llama3.2-3B-Instruct and Qwen3-4B-Base to demonstrate that ReLaX generalizes beyond specific foundation models.

As shown in Table 5, our method consistently outperforms the GRPO baseline across both types of base models. To further compare with token-level methods, we additionally include results from the recently released HICRA [38]. ReLaX achieves substantial gains on Minerva

Model	Size	MATH500	Minerva	AMC22	AMC23	AIME24	AIME25	Average
		Mean@1	*Mean@1*	*Mean@32*	*Mean@32*	*Mean@32*	*Mean@32*	
Llama3.2-Instruct	3B	46.8	15.4	16.1†	22.0	8.5	0	18.1
+Vanilla GRPO[32]	3B	55.4	22.8	26.4	40.0	16.3	1.4	27.1
+ReLaX (Ours)	3B	57.0	23.5	39.0	52.8	18.9	3.3	32.4
Δ (ReLaX - GRPO)	3B	+1.6	+0.7	+12.6	+12.8	+2.6	+1.9	+5.3
Qwen3-Base	4B	63.8	28.3	29.1†	38.9	9.4	5.3	29.1
+Vanilla GRPO[32]	4B	83.0	38.9	42.6	51.2	24.9	23.8	44.1
+HICRA [38]	4B	89.0	42.5	-	54.0	31.0	27.6	-
+ReLaX (Ours)	4B	90.2	48.5	52.6	64.5	30.9	27.6	52.3
Δ (ReLaX - GRPO)	4B	+6.2	+9.6	+10.0	+13.3	+6.0	+3.8	+8.2

Table 5. Supplemented comparison of LLM performance (mean@1 & mean@32) trained from Llama3.2-Instruct and Qwen3 across multiple text-only mathematical reasoning benchmarks. The performance gains of ReLaX over the GRPO baseline are highlighted in red. † indicates our reproduced results using publicly available models and standard evaluation code. "–" denotes missing results due to unavailable models.

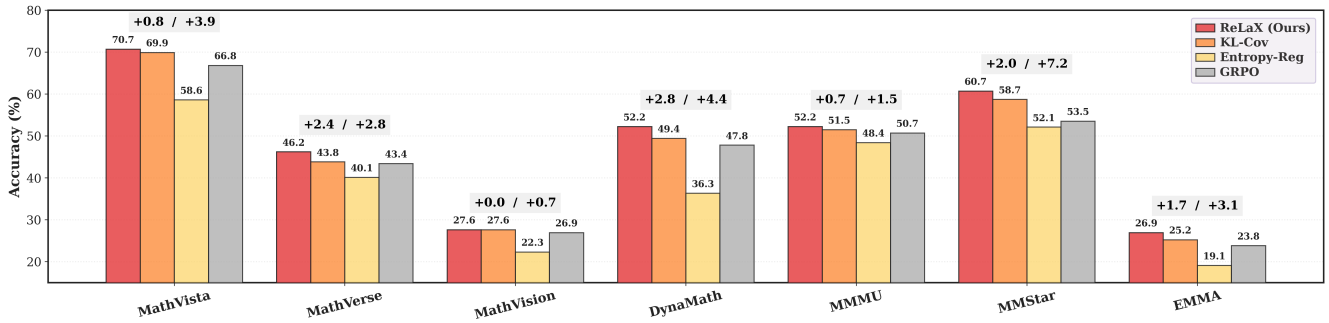


Figure 8. Complete comparison of multimodal benchmark performance for different training methods on Qwen2.5-3B-VL. This figure provides the extended results referenced in Fig. 7a. The values above the bars denote ReLaX's performance gains over KL-Cov (left) and vanilla GRPO (right).

and AMC, while remaining competitive on MATH500 and AIME. These results demonstrate that ReLaX generalizes effectively across different base models and maintains a stable advantage over token-level methods.

5.2. Computational Time Consumption

Compared with vanilla GRPO, ReLaX introduces additional computation for the model's latent representation. To assess its time efficiency, we provide a detailed runtime breakdown of the extra components, including Koopman dictionary learning and the actor update, and compare them against the vanilla GRPO. Following our ablation setup, this analysis is conducted on Qwen2.5-3B-VL and Qwen2.5-7B-Math.

As shown in Table 6, fitting the Koopman dictionary incurs only a small one-time cost (109 s for the 3B model and 132 s for the 7B model), which is negligible compared to the whole training process, as it is performed only once at initialization. During training, the primary additional overhead of ReLaX arises from computing the DSD score for each hidden state at every step, which increases the actor-update time by roughly 50% compared with vanilla GRPO. However, this component accounts for only about 10% of the total per-step runtime, making its impact on overall training time relatively minor. These observations indicate that the time overhead introduced by ReLaX is acceptable in practice, and we leave further optimization of its computational efficiency to future work.

5.3. More Details on Comparisons with Token-level Methods

To support the claims and conclusions presented in Sec. 4.4, we provide additional results from analytical experiments comparing ReLaX with token-level methods. The following section presents evaluation results across all multimodal reasoning benchmarks, as well as a detailed case study of the obtained LRMs.

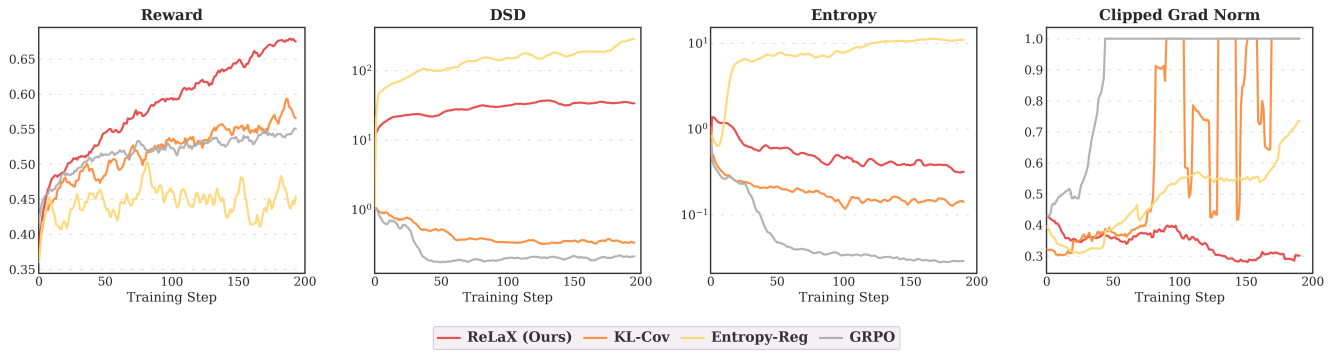


Figure 9. More training dynamics of vanilla GRPO, KL-Cov, Entropy Reg and our ReLaX on Qwen2.5-3B-VL. This figure provides the extended results for Fig. 7b and 7c.

Component	GRPO (s)	ReLaX (s)
Qwen2.5-3B-VL		
Fit Koopman dictionary	-	109
Update actor	490	578
Total / step	1052	1161 (+10.4%)
Qwen2.5-7B-Math		
Fit Koopman dictionary	-	132
Update actor	449	653
Total / step	2030	2273 (+12.0%)

Table 6. Comparison of the runtime breakdown per training step between vanilla GRPO and the proposed ReLaX on Qwen2.5-3B-VL and Qwen2.5-7B-Math.

5.3.1. Extra Evaluation Results of VLMs

Firstly, we provide additional results of token-level methods on 3B-VL models evaluated across all multimodal benchmarks. In addition to the results on MathVista, DynaMath, MMStar, and the two multidisciplinary subsets of EMMA reported in Fig. 7a, we further include comparisons on MathVerse, MathVision, MMMU, and the full EMMA benchmark in Fig. 8. These results reinforce that while ReLaX consistently outperforms the vanilla GRPO baseline, its main advantage over the token-level method, KL-Cov, appears in the multidisciplinary multimodal benchmarks, which rely more on visual information, rather than in the mathematics-dominant benchmarks where text plays a central role. Additional training dynamics of the reward and gradient norm are presented in Fig. 9. We observe that ReLaX exhibits a notably stable gradient norm, which is beneficial for optimization.

5.3.2. Case Study for Multimodal Reasoning

To investigate the behavior of models trained with the proposed ReLaX in multimodal reasoning, we present a detailed case study comparing ReLaX and KL-Cov [10] on 3B-VL models. We select a query from DynaMath [56], a recent benchmark designed to evaluate reasoning robustness by testing how well models perform under different variants of the same question, such as changes in visual numerical values. As shown in Table 7, the question asks for the perimeter of a rectangular prism, and the three variants differ only in the numerical values of length, width, and height. ReLaX-3B-VL consistently follows the correct formula, $P = 4(L + W + H)$, and produces accurate answers across all variants. In contrast, the model trained with KL-Cov shows clear sensitivity to these variations. Specifically, in the second variant, it fails to extract the height from the image and incorrectly treats the rectangular prism as a flat rectangle. In the third variant, it applies an incorrect computation procedure for the perimeter. Additional support for ReLaX's accurate visual understanding in multidisciplinary reasoning tasks is presented in Table 8.

5.3.3. Case Study for Text-only Reasoning

We further analyze the qualitative results on AMC23 (Tab. 9), comparing two 7B models fine-tuned from Qwen2.5-Math: one trained with ReLaX and the other with R1-zero-Div [47]. We examine two representative cases in which both models produce the correct final answer. Notably, both models engage in self-examination after generating the initial solution; however, this behavior reveals a fundamental difference in the quality of their reasoning. The model trained by the proposed ReLaX re-examines its results by applying the law of cosines, a mathematically grounded verification step directly relevant to the problem context. In contrast, R1-zero-Div attempts to validate its answer by generating Python code—despite lacking any execution environment or interpreter. This constitutes a form of hallucination induced by coarse-grained entropy encouragement: increasing token-level entropy may elevate surface-level diversity but offers no guarantee that the generated content remains meaningful.

Question: Given a rectangular prism with length l, width w, and height h, calculate its perimeter.

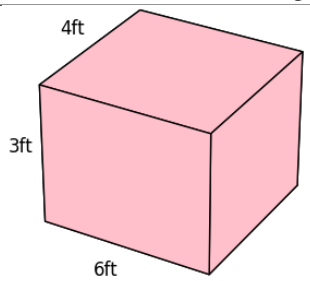
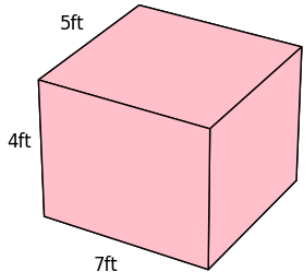
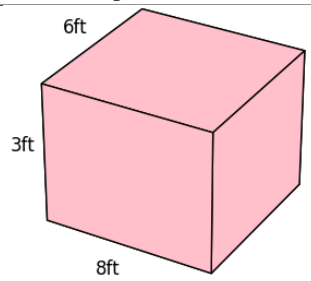
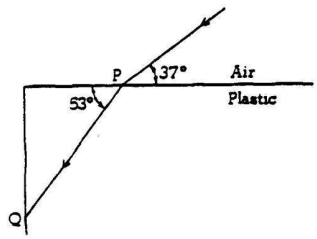
 Question Variant 1	 Question Variant 2	 Question Variant 3
ReLaX: To find the perimeter of a rectangular prism, ... Given: - Length (L) = 6 ft - Width (W) = 3 ft - Height (H) = 4 ft The formula for the perimeter P of a rectangular prism is: $P = 4(L + W + H)$ Substituting the given values into the formula: $P = 4(6 + 3 + 4)$ $P = 4(13)$ $P = 52$ So, the perimeter of the rectangular prism is $\boxed{52}$ feet. **[CORRECT]**	**ReLaX:** To find the perimeter of a rectangular prism, ... Given: - Length (L) = 7 ft - Width (W) = 4 ft - Height (H) = 5 ft The formula for the perimeter P of a rectangular prism is: $P = 4(L + W + H)$ Substituting the given values into the formula: $P = 4(7 + 4 + 5)$ $P = 4(16)$ $P = 64$ So, the perimeter of the rectangular prism is $\boxed{64}$ feet. **[CORRECT]**	**ReLaX:** To find the perimeter of a rectangular prism, ... Given: - Length (L) = 8 ft - Width (W) = 3 ft - Height (H) = 6 ft The formula for the perimeter P of a rectangular prism is: $P = 4(L + W + H)$ Substituting the given values into the formula: $P = 4(8 + 3 + 6)$ $P = 4(17)$ $P = 68$ So, the perimeter of the rectangular prism is $\boxed{68}$ feet. **[CORRECT]**
KL-Cov: To find the perimeter of the rectangular prism, ... The formula for the perimeter of the edges of a rectangular prism is: $P = 4(l + w + h)$ where l, w, and h are the length, width, and height of the prism, respectively. Substituting the given dimensions (6 feet, 3 feet, 4 feet): $P = 4(6+3+4) = 4 \times 13 = 52$ feet However, the problem seems to be asking for the perimeter of the faces, which is not a standard geometric term either. Given the dimensions provided, if we assume the problem is asking for the perimeter of the faces that would form a net of the prism, we need to consider the perimeter of the faces, ... Therefore, the correct answer is: $\boxed{52}$ **[CORRECT]**	**KL-Cov:** To find the perimeter of the rectangular prism, ... **The dimensions given for the base of the rectangular prism are 7 feet and 4 feet.** The perimeter P of a rectangle is calculated using the formula: $P = 2 \times (length + width)$ Here, the length is 7 feet and the width is 4 feet. Substituting these values into the formula, we get: $P = 2 \times (7+4) = 2 \times 11 = 22$ feet So, the perimeter of the rectangular prism (interpreted as the perimeter of the base) is: $\boxed{22}$ **[WRONG]**	**KL-Cov:** To find the perimeter of the rectangular prism, **we need to calculate the perimeter of each pair of opposite faces and then sum them up**. The perimeter of each pair of opposite faces is calculated as follows: 1. For the faces with dimensions 8 feet by 3 feet, the perimeter of one face is $2 \times (8 + 3) = 2 \times 11 = 22$ feet. Since there are two such faces, the total contribution to the perimeter from these faces is $2 \times 22 = 44$ feet. 2. ..., the total contribution to the perimeter from these faces is $2 \times 28 = 56$ feet. 3. ..., the total contribution to the perimeter from these faces is $2 \times 18 = 36$ feet. Adding these contributions together gives us the total perimeter of the rectangular prism: $44 + 56 + 36 = 136$ Therefore, the perimeter of the rectangular prism is $\boxed{136}$. **[WRONG]**

Table 7. Comparison of model responses on DynaMath between ReLaX and KL-Cov using three query variants with modified visual content, all paired with the same text prompt: "**Find the perimeter of the rectangular prism.**" Two typical reasoning failures caused by KL-Cov are highlighted in red and purple, respectively, while the correct counterparts by ReLaX are highlighted in green.

Question: A light ray enters a block of plastic and travels along the path shown. By considering the behavior of the ray at point P, determine the speed of light in the plastic. Options: A. 0.44 ;B. 0.88; C. 1.13; D. 2.26.



ReLaX: To determine the speed of light in the plastic, we need to use Snell's Law,...

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

Given:
- The angle of incidence $\theta_1 = 53°$.
- The angle of refraction $\theta_2 = 37°$.

We know that the index of refraction for air is approximately $n_1 = 1$. We need to find the index of refraction for the plastic, n_2.
Using Snell's Law:

$$1 \cdot \sin(53°) = n_2 \cdot \sin(37°)$$

...
The speed of light in a medium is given by:

$$v = \frac{c}{n}$$

...
Therefore, the speed of light in the plastic is:

$$v = \frac{3 \times 10^8 \text{ m/s}}{1.33} \approx 2.26 \times 10^8 \text{ m/s}$$

Hence, the correct answer is:

$$\boxed{D}$$

[CORRECT]

KL-Cov: To solve this problem, we need to use the principles of refraction and Snell's Law. We are given the angle of incidence at point P as $37°$ and the angle of refraction inside the plastic as $53°$.
First, let's recall Snell's Law, which states:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

... The refractive index of the plastic is $n_2 = 0.75$. The speed of light in a medium is given by:

$$v = \frac{c}{n}$$

However, we need to check the options provided. The options are in terms of a fraction of 10^8 m/s. The correct value for the refractive index of the plastic is actually $n_2 = \frac{3}{4} = 0.75$, but we need to re-evaluate the options given. The correct option based on the refractive index calculation is:

$$v = \frac{3 \times 10^8 \text{ m/s}}{0.75} = 4 \times 10^8 \text{ m/s} \approx 0.88 \times 10^8 \text{ m/s}$$

So the correct answer is:

$$\boxed{B}$$

[WRONG]

Table 8. Comparison of model responses on MMMU between ReLaX and KL-Cov. Failures of KL-Cov in correctly interpreting the visual input are highlighted in red, whereas the correct responses produced by ReLaX are highlighted in green.

Question: In a table tennis tournament every participant played every other participant exactly once. Although there were twice as many right-handed players as left-handed players, the number of games won by left-handed players was 40% more than the number of games won by right-handed players. (There were no ties and no ambidextrous players.) What is the total number of games played?

ReLaX: To solve the problem, we start by letting the side lengths of triangle ABC be a, b, and c with a being the smallest side and the sides in arithmetic progression. Since a is the smallest side and has a length of 6, we can let the side lengths be 6, $6+d$, and $6+2d$. Without loss of generality, we assume that the angle opposite the largest side is 120. Thus, the largest side is $6+2d$.

...

The area A of the triangle is given by:

$A = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{15(15-6)(15-10)(15-14)} = \sqrt{15 \cdot 9 \cdot 5 \cdot 1} = \sqrt{675} = 15\sqrt{3}$

However, we should verify our result using **the Law of Cosines** as well to ensure the correct angle is 120:

$c^2 = a^2 + b^2 - 2ab\cos(C)$

...

This confirms that our side lengths and angle are correct. Thus, the area of the triangle is: $15\sqrt{3}$

The problem specifies that the final answer should be in the form $m\sqrt{n}$ where m and n are positive integers and n has no square factors. Here, $m = 15$ and $n = 3$. Therefore, $m + n = 15 + 3 = 18$.

Thus, the final answer is: $\boxed{18}$

[CORRECT]

R1-zero-Div: Given that the side lengths of triangle ABC are in arithmetic progression and the smallest side has length 6, let's denote the side lengths as 6, $6+d$, and $6+2d$ for some positive number d. Without loss of generality, we can assume that the angle of 120 is opposite the largest side, which is $6+2d$.

By the Law of Cosines, we have:

$(6+2d)^2 = 6^2 + (6+d)^2 - 2 \cdot 6 \cdot (6+d) \cdot \cos(120)$

...

Thus, $m = 15$ and $n = 3$, and the final answer is $m + n = 15 + 3 = 18$. Let's confirm this using **Python and SymPy**.

```python
import sympy as sp
...
print(result)
```

Output: '18'

The area of triangle ABC is $15\sqrt{3}$. Therefore, $m = 15$ and $n = 3$, and the final answer is $m + n = 18$.

Thus, the final answer is $\boxed{18}$.

[CORRECT]

Table 9. Comparison of model responses on AMC23 between ReLaX and R1-zero-Div. The self-examination behaviors exhibited during reasoning are highlighted in green and red, respectively.